Hariton Pitsik

JUNIOR ML ENGINEER · ML INTERN
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Education

Saratov State University — Saratov, Russia
BACHELOR | FUNDAMENTAL COMPUTER SCIENCE AND INFORMATION TECHNOLOGY — September 2023 - June 2027

Skills

Programming languages	Python, Rust, SQL
Spoken languages	Russian (native), English (B2)
Technologies	Git & GitHub, Docker, Linux, LaTeX, Typst, Jupyter, NumPy, Pandas, Matplotlib & Seaborn, Scikit-learn, XGBoost, CatBoost, FastAPI, ClearML, Streamlit, PyTorch, Transformers, PostgreSQL
Concepts	Mathematics & Statistics (Linear Algebra, Calculus, Probability Theory), Classical ML (Supervised & Unsupervised Learning), Deep Learning, Natural Language Processing & LLM, Computer Vision

Some achievements

Teaching & Social

- Participant of Development Students Club (DSC)
- Teach machine learning as part of a DSC community.
  - Lecture recordings on NumPy (YouTube) and Pandas (YouTube).
  - Machine Learning Club playlist: (YouTube)

Practical Experience

Speech-to-Text System for Dialogues and Catalog Terms — haritonn/asr_practice

PRODUCTION INTERNSHIP PROJECT.

- Developed a speech-to-text system for Russian-language dialogues, combining Faster-Whisper transcription, Silero VAD, and pyannote speaker diarization;
- Implemented recognition of domain-specific catalog terms with a NeMo CTC context graph, eliminating the need to fine-tune the ASR model;
- Linked transcripts and detected terms to dialogue participants; evaluated solution quality using WER, CER, DER, and terminology F1.

Projects

Local RAG System — haritonn/pdf_rag

- Built a fully local RAG assistant that answers questions over uploaded PDF documents;
- Made retrieval configurable from the UI: switchable LLM, embedding model, vector database, and Top-K sources;
- Made answers auditable by exposing source documents and retrieved contexts in the Streamlit interface; implemented a modular architecture with Qdrant.

Transformers from Scratch — haritonn/transformers-scratch

- Implemented a Transformer encoder-decoder architecture from scratch in PyTorch;

- Built a modular training pipeline for seq2seq tasks;
- Added flexible configuration of training and experiments.

Caption Generator
haritonn/caption_gen

- Engineered an end-to-end image-captioning pipeline on Flickr8k: ResNet-50 encoder + soft-attention LSTM decoder in PyTorch;
- Applied scheduled sampling, label smoothing, attention regularization, gradient clipping, and early stopping to improve training stability;
- Built a reproducible experiment workflow with deterministic train/validation/test splits, checkpointing, and optional ClearML tracking;
- Evaluated caption quality with BLEU and METEOR; reached 0.1655 BLEU and 0.3774 METEOR on the validation split (37m. weights).

cargo-smi — NVIDIA GPU Monitor
haritonn/cargo-smi

- Developed a responsive terminal dashboard in Rust for unified monitoring of NVIDIA GPUs and host-system resources;
- Integrated NVML to display per-GPU temperature, utilization, VRAM, CUDA/driver versions, and GPU-consuming processes;
- Added multi-GPU navigation, configurable auto-refresh, manual refresh, and a 120-sample utilization history chart;
- Surfaced CPU, RAM, swap, and the 20 most CPU-intensive processes in the same TUI using Ratatui, Crossterm, and Sysinfo.

ResNet with Attention
haritonn/resnet_attention

- Implemented baseline ResNet-50 and a channel-attention variant from scratch in PyTorch;
- Improved classification quality over the baseline by adding channel attention; compared convergence, accuracy, and F1 score;
- Built a reproducible training and inference workflow with early stopping, attention visualizations, and ClearML experiment tracking.

Coursework

2nd Year: Generative Computer Vision
haritonn/coursework2

Compared diffusion models, GANs, and large multimodal model (LMM)-based approaches for virtual try-on. Evaluated visual quality and inference time on a custom dataset.

3rd Year: Text-Embedding Geometry & HDBSCAN Clustering
haritonn/coursework3

Investigated how embedding geometry affects news-event clustering: ran 2,604 HDBSCAN experiments across 3 embedding families, 93 feature spaces, and 28 parameter values. Showed that UMAP improved the best F1 for every embedding family; achieved F1 = 0.837 with Jina Embeddings v5 + UMAP(10).